Orthogonality

Orthogonal Vectors

Theorem: 1

The inner product $x^T y$ is zero if and only if x and y are orthogonal vectors.

Remarks:

If $x^T y > 0$ angle between x and y is acute.

If $x^T y < 0$ angle between x and y is greater than $90°$.

Property – Mutually orthogonal vectors

if non-zero vectors are mutually orthogonal then those vectors are linearly independent.

if non-zero vectors $v_1, v_2, \dots, v_k$ are mutually orthogonal then those vectors are linearly independent.

Orthogonal subspaces

Two subspaces are V and W of the same space $\mathbb{R}^n$ are orthogonal if every vector in V orthogonal to every vector in W.

Remarks:

- The subspace $\{0\}$ is orthogonal to all subspaces.
- A line can be orthogonal to another line.
- A line can be orthogonal to a plane.
- A plane cannot be orthogonal to a plane.

Problem

Verify whether the plane spanned by $v_1 = (1,0,0,0)$ and $v_2 = (1,1,0,0)$ is orthogonal to the line spanned by $w = (0,0,4,5)$.

Fundamental theorem of Orthogonality

- The row space is orthogonal to the nullspace.
- The column space is orthogonal to the left nullspace.

Remarks:

Given the matrix $A_{m\times n}$:

- The dimension of row space and the dimension of null space add to n.
- The dimension of column space and the dimension of the left null space add to m.

Orthogonal complement

Given a space V of $\mathbb{R}^n$, the space of all vectors orthogonal to V is called the orthogonal complement of V. It is denoted by $V^{\perp}$.

Remarks:

- The null space is the orthogonal complement of the row space.
- The left null space and the column space are orthogonal complements.
- Two subspaces V and W can be orthogonal without being orthogonal complements.

Theorem

Every vector b in the column space comes from exactly one vector x_r in the row space.

Fundamental Theorem of Linear Algebra - II

- The null space is the orthogonal complement of the row space in $\mathbb{R}^n$.
- The left null space is the orthogonal complement of the column space in $\mathbb{R}^m$.